

Ganeshram.M

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CSA0925

```
import java.awt.*;
import java.awt.event.*;

public class Main extends Frame implements ActionListener {

    Label l1, l2, l3, l4, l5, l6, l7;
    TextField t1, t2, t5, t6;
    Choice department, designation;
    CheckboxGroup performance;
    Checkbox excellent, good, average, poor;
    Button generate, reset;
    TextArea output;

    Main() {
        setTitle("Employee Performance & Payroll Dashboard");
        setSize(750, 650);
        setLayout(new GridBagLayout());

        GridBagConstraints gbc = new GridBagConstraints();
        gbc.insets = new Insets(8, 8, 8, 8);
        gbc.fill = GridBagConstraints.HORIZONTAL;

        l1 = new Label("Employee ID:");
        l2 = new Label("Employee Name:");
        l3 = new Label("Department:");
        l4 = new Label("Designation:");
        l5 = new Label("Basic Salary:");
        l6 = new Label("Working Days:");
        l7 = new Label("Performance Rating:");

        t1 = new TextField(20);
        t2 = new TextField(20);
        t5 = new TextField(20);
        t6 = new TextField(20);

        department = new Choice();
```

```
department.add("IT");
department.add("HR");
department.add("Finance");
department.add("Marketing");
department.add("Sales");
```

```
designation = new Choice();
designation.add("Developer");
designation.add("Manager");
designation.add("Tester");
designation.add("Analyst");
designation.add("Team Lead");
```

```
performance = new CheckboxGroup();
```

```
excellent = new Checkbox("Excellent", performance, false);
good = new Checkbox("Good", performance, true);
average = new Checkbox("Average", performance, false);
poor = new Checkbox("Poor", performance, false);
```

```
Panel ratingPanel = new Panel(new FlowLayout(FlowLayout.LEFT));
ratingPanel.add(excellent);
ratingPanel.add(good);
ratingPanel.add(average);
ratingPanel.add(poor);
```

```
generate = new Button("Generate Report");
reset = new Button("Reset");
```

```
generate.addActionListener(this);
reset.addActionListener(this);
```

```
output = new TextArea(12, 55);
output.setEditable(false);
```

```
// Row 1
gbc.gridx = 0;
gbc.gridy = 0;
add(l1, gbc);
```

```
gbc.gridx = 1;  
add(t1, gbc);
```

```
// Row 2  
gbc.gridx = 0;  
gbc.gridy = 1;  
add(l2, gbc);
```

```
gbc.gridx = 1;  
add(t2, gbc);
```

```
// Row 3  
gbc.gridx = 0;  
gbc.gridy = 2;  
add(l3, gbc);
```

```
gbc.gridx = 1;  
add(department, gbc);
```

```
// Row 4  
gbc.gridx = 0;  
gbc.gridy = 3;  
add(l4, gbc);
```

```
gbc.gridx = 1;  
add(designation, gbc);
```

```
// Row 5  
gbc.gridx = 0;  
gbc.gridy = 4;  
add(l5, gbc);
```

```
gbc.gridx = 1;  
add(t5, gbc);
```

```
// Row 6  
gbc.gridx = 0;  
gbc.gridy = 5;  
add(l6, gbc);
```

```

gbc.gridx = 1;
add(t6, gbc);

// Row 7
gbc.gridx = 0;
gbc.gridy = 6;
add(l7, gbc);

gbc.gridx = 1;
add(ratingPanel, gbc);

// Buttons
gbc.gridx = 0;
gbc.gridy = 7;
add(generate, gbc);

gbc.gridx = 1;
add(reset, gbc);

// Output
gbc.gridx = 0;
gbc.gridy = 8;
gbc.gridwidth = 2;
gbc.fill = GridBagConstraints.BOTH;
add(output, gbc);

setVisible(true);

addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        dispose();
    }
});
}

public void actionPerformed(ActionEvent e) {

    if (e.getSource() == reset) {
        t1.setText("");
        t2.setText("");
    }
}

```

```

t5.setText("");
t6.setText("");
department.select(0);
designation.select(0);
performance.setSelectedCheckbox(good);
output.setText("");
}

if (e.getSource() == generate) {

    String id = t1.getText().trim();
    String name = t2.getText().trim();

    if (id.isEmpty() || name.isEmpty()) {
        output.setText("Error: Employee ID and Employee Name cannot be empty.");
        return;
    }

    double basicSalary;
    int workingDays;

    try {
        basicSalary = Double.parseDouble(t5.getText().trim());
        workingDays = Integer.parseInt(t6.getText().trim());
    } catch (Exception ex) {
        output.setText("Error: Enter valid numeric values for salary and working days.");
        return;
    }

    if (basicSalary <= 0) {
        output.setText("Error: Basic salary must be greater than 0.");
        return;
    }

    if (workingDays < 0 || workingDays > 31) {
        output.setText("Error: Working days must be between 0 and 31.");
        return;
    }

    String rating = performance.getSelectedCheckbox().getLabel();

```

```

// Salary calculation
double attendanceSalary = basicSalary * workingDays / 26.0;

double hra = attendanceSalary * 0.20;
double da = attendanceSalary * 0.10;

double grossSalary = attendanceSalary + hra + da;

double bonusRate;

if (rating.equals("Excellent")) {
    bonusRate = 0.15;
} else if (rating.equals("Good")) {
    bonusRate = 0.10;
} else if (rating.equals("Average")) {
    bonusRate = 0.05;
} else {
    bonusRate = 0.00;
}

double performanceBonus = grossSalary * bonusRate;
double netSalary = grossSalary + performanceBonus;

output.setText("");

output.append("=====\n");
output.append("    EMPLOYEE PAYROLL REPORT\n");
output.append("=====\n\n");

output.append("Employee ID    : " + id + "\n");
output.append("Employee Name  : " + name + "\n");
output.append("Department    : " + department.getSelectedItem() + "\n");
output.append("Designation   : " + designation.getSelectedItem() + "\n");
output.append("Basic Salary   : " + String.format("%.2f", basicSalary) + "\n");
output.append("Working Days   : " + workingDays + "\n");
output.append("Performance    : " + rating + "\n\n");

output.append("-----\n");
output.append("Attendance Salary : " + String.format("%.2f", attendanceSalary) + "\n");

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```

        output.append("HRA (20%)      : " + String.format("%.2f", hra) + "\n");
        output.append("DA (10%)       : " + String.format("%.2f", da) + "\n");
        output.append("Gross Salary   : " + String.format("%.2f", grossSalary) + "\n");
        output.append("Performance Bonus : " + String.format("%.2f", performanceBonus) +
"\n");
        output.append("Net Salary      : " + String.format("%.2f", netSalary) + "\n");
        output.append("-----\n");
        output.append("      Payroll Generated Successfully\n");
        output.append("=====\n");
    }
}

public static void main(String[] args) {
    new Main();
}
}

```

Employee ID:	1922
Employee Name:	Ganeshram
Department:	IT
Designation:	Developer
Basic Salary:	40000
Working Days:	25
Performance Rating:	☐ Excellent ☒ Good ☐ Average ☐ Poor

Generate Report	Reset
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Employee ID:	1922
Employee Name:	Ganeshram
Department:	IT
Designation:	Developer
Basic Salary:	40000
Working Days:	25

Performance Rating: ☐ Excellent ☒ Good ☐ Average ☐ Poor

Generate Report	Reset
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Performance	: Good
Attendance Salary	: 38461.54
HRA (20%)	: 7692.31
DA (10%)	: 3846.15
Gross Salary	: 50000.00
Performance Bonus	: 5000.00
Net Salary	: 55000.00
Payroll Generated Successfully	
=====	